CSC-301

Assignment-2

Name - Aditya Pratap Singh Bhadoria
Enroll - 24114005

Q.1]

i) Attribute - Descriptive properties possessed by each member of an entity set.

ii) Domain - The set of possible values associated with an attribute

iii) Entity - A thing or object in the real world that is distinguishable from all other objects.

iv) Relationship - An association among two or more entities.

v) Entity set - A set of entities of the same type that share the same properties, or attributes.

vi) Relationship set - A collection of similar relations (of same type)

vii) One-to-many relationship - A mapping cardinality where an entity in the first set is associated with several entities of second set, but from the second set is associated with atmost one from one entity in first set.

viii) Many-to-many -"- - An entity from first set associated with several entities of 2nd set, and vice versa.

ix) Participation constraint - Expresses whether every entity in an entity set must participate in a relationship (total participation) or if only some entities participate (partial participation)

x) Overlap constraint - Specifies whether a single entity in a superclass can simultaneously belong to more than 1 subclass.

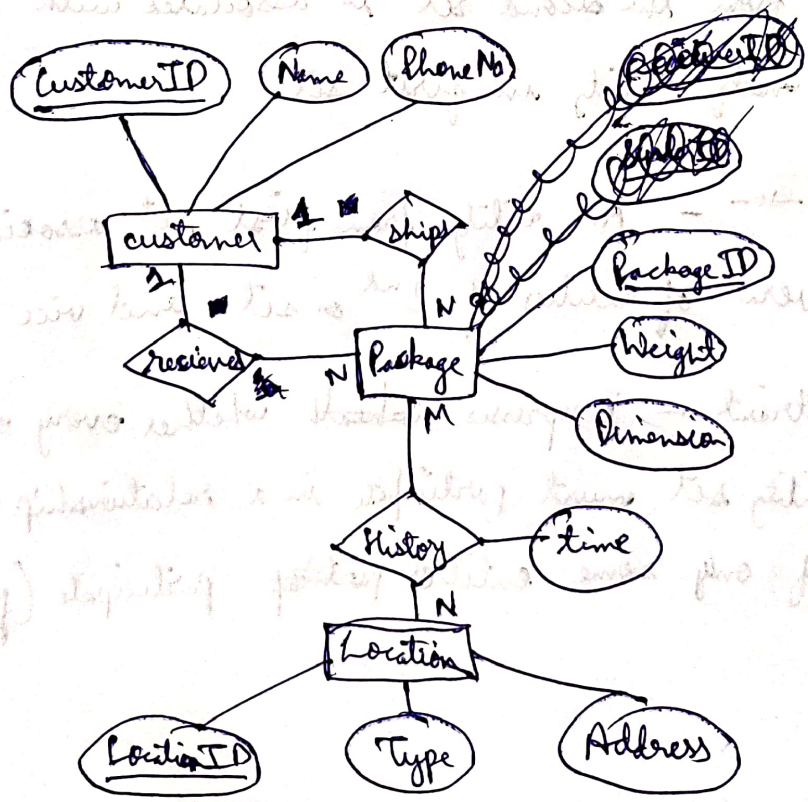
xi) Covering constraint - Specifies whether a single entity in a superclass must belong to at least one of its subclasses.

xii) Aggregation - An ER modelling abstraction that treats an entire relationship set and its participating entities as a single higher-level entity, allowing it to participate in other relationships.

xiii) Weak entity set - An entity type that does not have key attributes of its own and whose entities are identified by being related to specific entities from another strong entity type.

xiv) Role indicator - Labels concatenated with an attribute name to provide a unique name for each attribute when an entity set plays more than one role in a single relationship.

Q2) Worldwide delivery company ER diagram -



Relationship schemas & constraints →

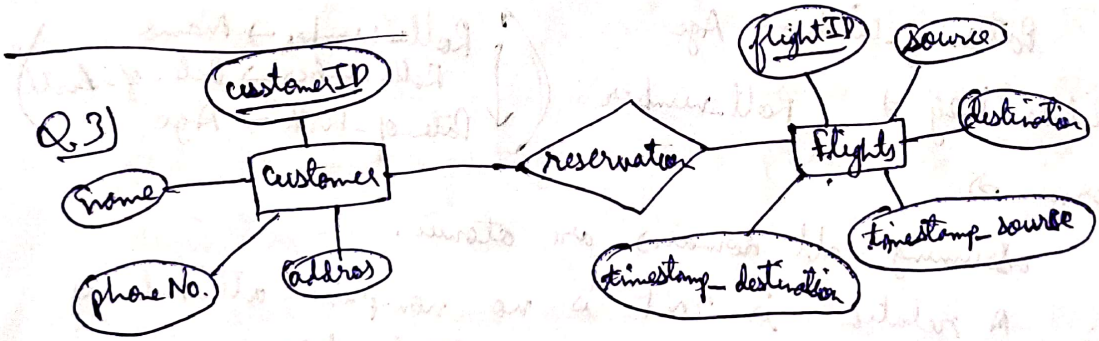
1) Customer (CustomerID , Name , Phone)
   → Primary key - CustomerID

2) Package (PackageID , senderID , ReceiverID, Weight , dimensions)
   → Primary key - PackageID
   → Foreign key - senderID references Customer (CustomerID)
   → " " - ReceiverID " " (")

3) Location (LocationID , type , Address)
   → Primary - Location ID

4) ~~Tracking~~ History (PackageID, LocationID, time)
   → Primary Key - PackageID, LocationID, Time (combined to ensure uniqueness per location scan)
   → Foreign key - PackageID references Package (PackageID)
   → Foreign key - LocationID references Location (Location ID)

Q.3)



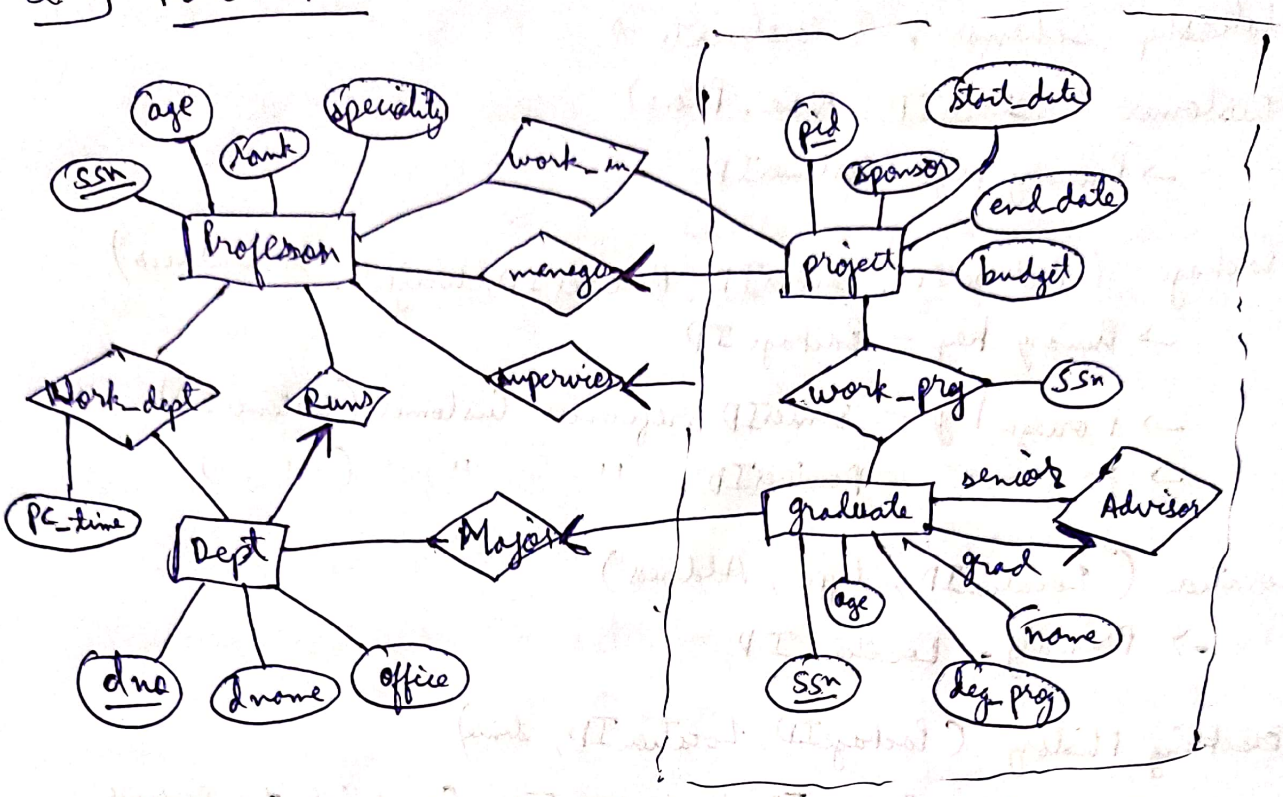
Relation schemas :-

1) Customer (customerID , name , phoneNo , address)
   → Primary - customerID

2) Flights (flightID, source, destination, timestamp_source, timestamp_destination)
   → Primary key - flightID

Scanned with OKEN Scanner

Q4) Part-B



Q5) For R = (Roll_Number, Name, Date_of_birth, Age)

the FD are :- (applicable)

Roll_number → Name

Roll_number → Date_of_birth

Date_of_birth → Age

So, candidate key ≠ Roll_number (Roll_number → Name, Roll_number → Date_of_birth, Date_of_birth → Age)

Normal form =>

i) 1NF: assuming all domains are atomic.

ii) 2NF: A relation is 2NF if no non-prime attribute is partially dependent on any candidate key.

As (Roll_number) is a single attribute, partial dependency is impossible. So, R is in 2NF.

iii) 3NF: if for every non-trivial FD X → A, either X is superkey or A is prime attribute.

For Date_of_birth → Age, neither Date_of_birth is superkey nor Age is a prime attribute.

So, R is in 2NF.

Q.6] Class Definition - A relation is 3NF if for every FD $X \to A$, either X is superkey or A is prime attribute.

Ques definition - A relation is 3NF if there are no non-prime attributes A such that are transitively dependent on a key.

Transitive dependency definition $\Rightarrow \alpha \to \beta$, $\beta \not\to \alpha$, and $\beta \to A$ (where $A \notin \alpha \cup \beta$)

Proof:

1) Assume a relation violates the class definition: There exists an FD $X \to A$, where X is not a superkey and A is a non-prime attribute. Let K be a candidate key, so $K \to X$ holds. Because X is not a superkey, $X \not\to K$ holds. As $X \to A$ hold and A is a non-prime, the sequence $K \to X \to A$ matches the definition of a transitive dependency. So, this violates new condition as well.

2) Assume new definition violation: This is a transitive dependency $K \to X$, $X \not\to K$ and $X \to A$, where A is non-prime. Looking at the FD $X \to A$: X can't be a superkey as $X \not\to K$. A is given as a non-prime attribute, which violates the condition that X is a superkey or A is prime. So, it violates class definition.

As violation of one guarantees violation of another, these statements are equivalent.

Q.7] 1) Every 3NF is in 2NF:

As 3NF, every non-trivial FD $X \to A$, either X is superkey or A is a prime attribute.

Assume it isn't in 2NF, means there is a non-prime attribute A that is partially dependant on on a key K.

So, there exist proper subset $y \subset K$, such that $y \to A$.

Scanned with OKEN Scanner

~~Example to FD y → A~~, Now $y \rightarrow A$

As y is a proper subset, it isn't a superkey, and as A is non-prime, it isn't a prime attributes, both violating 3NF conditions, which contradicts our assumption.

So, any 3NF relation is in 2NF.

2) Every partial dependency is a transitive dependency:

Let attribute A be partially dependent on key K, implies there is proper subset $y \subset K$, such that $y \rightarrow A$.

Mapping to a transitive dependency formula: $(\alpha \rightarrow \beta, \beta \not\rightarrow \alpha, \beta \rightarrow A)$:

Let $\alpha = K$, $\beta = y$.

As $y \subset K$, $K \rightarrow y$ holds ($\alpha \rightarrow \beta$ holds)

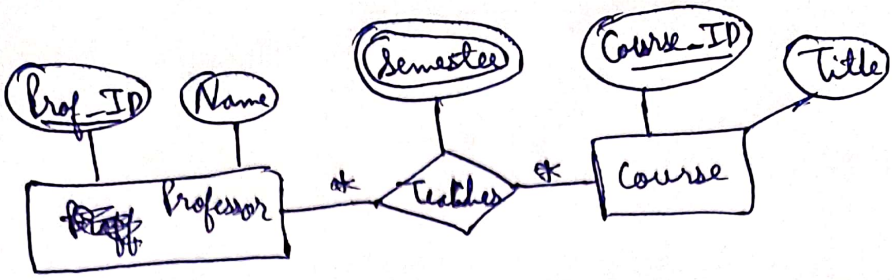
and as y is a proper subset of K, $y \not\rightarrow K$ ($\beta \not\rightarrow \alpha$)

By partial dependency, $y \rightarrow A$ holds, ($\beta \rightarrow A$),

So, $K \rightarrow y \rightarrow A$ satisfies all conditions of a transitive dependency. Every partial depency is transitive dependency.

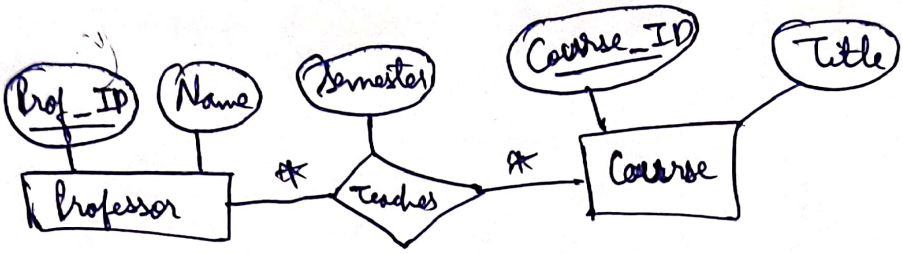
Q. 4] Part A

1)



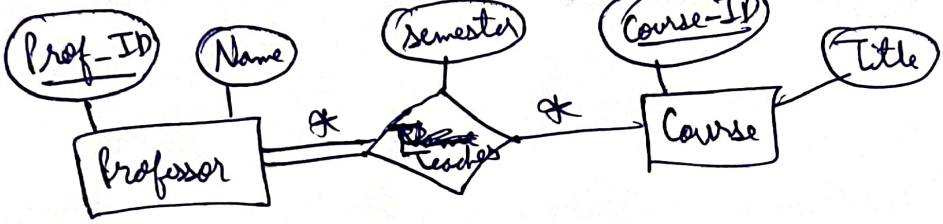
(several sem recorded)

2)



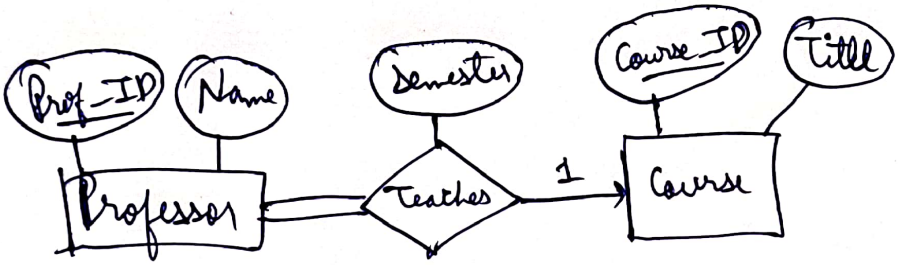
(latest sem recorded)

3)



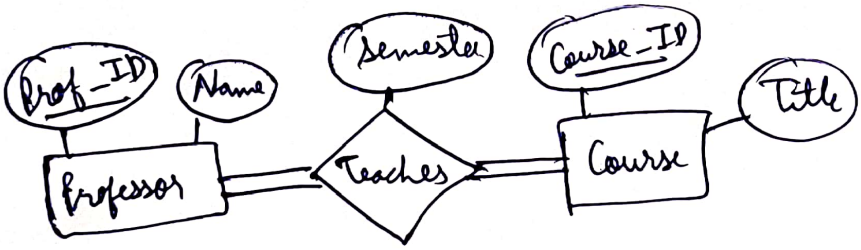
(every prof must teach)

4)



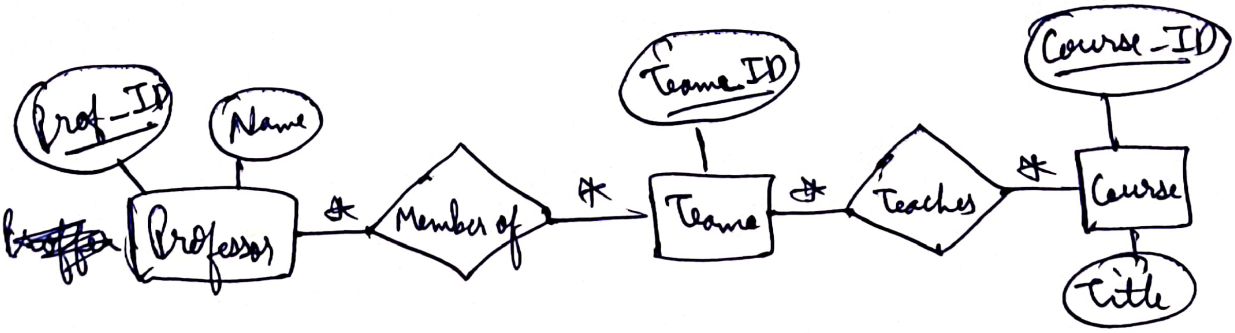
(every prof exactly 1 course)

5)



(" every course must be taught)

6)



Scanned with OKEN Scanner